

SEICHE

A funding-stress and leveraged-positioning early-warning terminal
for US money markets and the Treasury capital-market complex

*seiche /seɪʃ/ n. — a standing wave that oscillates in an enclosed body of water, often imperceptible from the shore until it sloshes over the edge.
Funding stress behaves the same way: it builds inside the system's plumbing while price screens stay calm.*

Product report and technical documentation

Version 0.1 · Built and live-verified 2026-07-06

Author: Mrinal (with Claude) · Side project

Repository: `~/dev/seiche` (local, private)

Data cost: \$0 · All sources are free, keyless, official public APIs

1 · Executive summary

What it is. Seiche is a single-pane analytics terminal that watches the machinery underneath US dollar markets: bank reserves, the repo market, the Treasury General Account, the Fed's balance sheet, Treasury auctions, and the leveraged hedge-fund positioning that sits on top of all of it. It converts about forty raw official data series into seven computed "engines" and one composite score, the **Seiche Index**, with a plain-language regime call: CALM, EROSION, STRAIN, or STRESS.

Why it exists. Every notable market stress event of 2025 and 2026 announced itself first in the plumbing, not in prices. The September 15, 2025 tax-date squeeze, the record Standing Repo Facility draws of October 31 (\$50B) and December 31, 2025 (\$74.6B), and the April 2025 basis-trade unwind were all preceded by measurable moves in repo-rate distribution tails, facility usage, and reserve arithmetic while equity and bond screens looked normal. The data that would have warned you is public and free. It is just scattered across five agencies, published in different formats at different lags, and never synthesized. Seiche is that missing synthesis layer.

Who it is for. Rates and repo desks, macro and relative-value hedge funds, treasury teams, and serious independent macro analysts. The positioning line: *the opinionated funding-stress terminal for people who cannot expense a \$32,000 Bloomberg seat.*

Current reading (2026-07-06). Seiche Index 39 ·

EROSION

Structurally tight, acutely calm. The overnight reverse-repo buffer is effectively gone (\$2B against a \$2.55T peak), reserves sit at \$2,967B, below the model's estimated scarcity kink of roughly \$3,660B, and the forward reserve path flags July 31 and August 31 as crunch-risk dates. At the same time the acute indicators are quiet: rate-distribution tails are normal, SOFR prints 1bp below the interest rate on reserves, and Standing Repo Facility take-up is zero. "Erosion, not strain" is exactly what an honest reading of this market looks like.

2 · Is the data real? Verification against primary sources

Yes. Every input is pulled live from the official API of the agency that produces the statistic. Nothing is scraped from third parties, estimated from news, or synthetic. The table below shows the dashboard's headline values re-verified with fresh, independent API calls made directly to each agency on 2026-07-06, outside Seiche's own cache.

Metric	Seiche shows	Fresh pull from primary source	Source of record	Match
SOFR (secured overnight financing rate)	3.64% · 2026-07-02	3.64%, P99 3.72%, \$3,208B volume	NY Fed markets API	✓
EFFR (effective fed funds rate)	3.63% · 2026-07-03	3.63%	NY Fed / FRED	✓
IORB (interest on reserve balances)	3.65% · 2026-07-06	3.65%	Federal Reserve via FRED	✓
Bank reserves	\$2,967B · 2026-07-01	\$2,966.897B (WRESBAL)	Fed H.4.1 via FRED	✓
ON RRP take-up	\$2B · 2026-07-02	\$2.175B	NY Fed via FRED	✓
Treasury General Account	\$807B · 2026-07-02	\$807.4B opening balance	Daily Treasury Statement, FiscalData API	✓
Standing Repo Facility take-up	\$0.00B · 2026-07-06	\$0.0B accepted at 3.75% offering rate	NY Fed repo operations API	✓
Leveraged-fund UST futures shorts	\$1,060B total (6 contracts)	e.g. 10Y note: 2,267,559 contracts short as of 2026-06-23	CFTC Traders in Financial Futures	✓

What is real data and what is a model output

The **inputs** above are real official statistics. The **engines** (kink estimate, forward reserve path, echo similarity, RV size proxy, digestion index, the Seiche Index itself) are computed opinions built from those inputs. Each engine publishes its method string on the dashboard, and each model carries its own confidence discount. The tool never presents a model output as an observation. Known honest lags: CFTC positioning is published Friday for the prior Tuesday (T+3 by construction); OFR repo series marked preliminary are T+1; reserves and the Fed balance sheet are weekly (Wednesday). The dashboard shows the as-of date on every number, and the July 4 holiday weekend explains the 2026-07-02 stamps at the time of writing.

3 · Why now: the market regime this tool was built for

Three research sweeps (money-market plumbing, capital-market themes, competitive landscape) were run before a line of code was written. The verdict, in numbers:

- **Quantitative tightening ended December 1, 2025.** The Fed's balance sheet fell from a \$8.97T peak to roughly \$6.7T, then the Fed pivoted to Reserve Management Purchases of about \$40B of T-bills per month. It stopped shrinking because the plumbing told it to.
- **The shock absorber is empty.** The overnight reverse-repo facility, \$2.55T at its 2022 peak, now holds about \$2B. Until 2025, Treasury issuance drained the RRP first and reserves were insulated. Now every Treasury settlement and every tax date hits bank reserves directly, dollar for dollar.
- **Reserves are testing the "ample" floor.** Roughly \$2.97T, about 9.6% of GDP. The Q4 2025 squeezes (SOFR-IORB spreads at their widest since 2020, record SRF usage) happened at reserve levels only slightly below today's.
- **The leveraged Treasury complex is historically large.** Hedge-fund net repo borrowing reached roughly \$1.8T by end-2025. Published estimates of the basis trade disagree by half a trillion dollars (Morgan Stanley ~\$1.5T vs IMF ~\$1T) partly because methods are opaque. The swap-spread trade doubled to about \$631B in fifteen months. Regulators (IMF April 2026 GFSR, FSB February 2026, Dallas Fed May 2026) have all flagged it; Fed staff describe their own live tracker as experimental.
- **Every recent break started in the plumbing.** Sep 15, 2025: SOFR printed 18bp over EFFR on a tax date. Oct 31, 2025: \$50B SRF draw. Dec 31, 2025: \$74.6B SRF record with year-end repo at 4.0%. April 2025: tariff shock forced a basis unwind with 50bp intraday swings in the 10-year. In each case, distribution tails, facility usage, or settlement arithmetic moved first.

The competitive gap

Tier	Examples	What they have	What they lack
Institutional terminals	Bloomberg (\$31,980/yr/seat), LSEG, Macrobond, Exante	All the raw data, real time	No opinionated funding-stress synthesis; you assemble six screens yourself; no forward reserve projection
Official dashboards	OFR Short-Term Funding Monitor, NY Fed Data Hub, FiscalData	Authority, free APIs	Siloed, backward-looking, zero synthesis, no alerts, no forecasts
Research / newsletters	Wrightson ICAP, Fed Guy, Skyrn's repo commentary, Pozsar	The narrative and the judgment	Not a tool: no data product, no API, no live monitoring
Retail / open source	TradingView net-liquidity scripts, GitHub FRED wrappers	Free, opinionated	Single-formula depth; no positioning, no repo tails, no auction analytics, no provenance

No product in any tier fuses FRED + NY Fed + OFR + Treasury FiscalData + CFTC into one forward-looking, alerting-ready instrument. That fusion is Seiche's entire reason to exist, and because every source is free, the moat is the synthesis logic, not data licensing.

4 · The seven engines, in detail

4.1 Kink Engine — where does reserve scarcity actually start?

The demand curve for bank reserves is flat when reserves are abundant (the SOFR-IORB spread ignores reserve changes) and turns steeply upward when they get scarce. The point where it bends, the "kink", is the single most important unknown in money markets: the Fed publishes its own estimate (Reserve Demand Elasticity) only as occasional research. Seiche fits it continuously.

```
spread_bp = a + slope × max(0, kink_ratio - reserves/GDP) + ε  
kink_ratio found by grid search over 121 candidates; weekly data 2017-present; spreads  
winsorized at 1/99%
```

Current output: kink $\approx$ \$3,661B of reserves; actual reserves \$2,967B, so the system is operating about \$694B *below* the estimated bend, with fit R^2 0.62. Crucially, the engine also checks itself against the market: the fitted curve predicts a SOFR-IORB spread of -1bp at today's reserve level, and the market prints -2bp. That 0.91 consistency score multiplies the engine's contribution. When model and market disagree, the engine's voice is automatically quieted rather than trusted.

Interpretation: being below a shallow kink means reserve scarcity is real but early; the slope of the fitted scarcity region is still gentle. This matches the Fed's own behavior (it ended QT and began bill purchases precisely because elasticity turned negative).

4.2 Liquidity Weather — the six-week forward reserve forecast

With the RRP buffer at zero, reserves tomorrow are an arithmetic identity: today's reserves, plus Fed balance-sheet drift, minus whatever the Treasury General Account absorbs. The TGA moves in a strongly seasonal monthly rhythm (mid-month settlements, corporate tax dates on the 15th of March/April/June/September/December, month-end flows). Seiche learns that rhythm from about three years of Daily Treasury Statement history, projects it forward 42 business days, and adds honest uncertainty bands built from the 20th/80th percentiles of the model's own backtested cumulative errors.

Current output: the path drifts sideways around \$2.9-3.0T with wide bands. Two crunch windows are flagged: **July 31** and **August 31, 2026**, month-end dates where the worst-case band touches roughly \$2,831-2,837B while reserves already sit below the kink. A design subtlety: when reserves are already below the kink, "proximity to the kink" stops discriminating (every day would flag), so the engine collapses to flagging only calendar pressure dates. Degenerate signals are muted by design.

4.3 Tail Seismograph — the tails detach before the median moves

The NY Fed publishes the full daily distribution of secured funding rates, not just the median: the 1st, 25th, 75th, and 99th percentiles of SOFR, TGCR, and BGCR. Before any squeeze reaches the headline print, some desk somewhere pays up first, and it shows in the 99th percentile. September 15 and December 31, 2025 both exhibited tail detachment days before the median moved. Seiche computes P99 – P50 for each rate in basis points, z-scores it against a trailing year, blends the three rates (SOFR 0.5, TGCR 0.3, BGCR 0.2), and smooths with a 5-day EWMA.

Current output: tail index $z \approx 0$, SOFR tail 8bp over median, SOFR-IORB at -1bp. Quiet. This is the engine that will move first when the next squeeze builds, and its 24% weight is the largest in the composite for exactly that reason.

4.4 Echo Engine — does today rhyme with any pre-crisis run-up?

Seiche maintains a daily state vector of seven plumbing z-scores (SOFR-IORB, EFFR-IORB, BGCR-SOFR, RRP level, 5-day TGA change, 4-week reserve change, SRF usage). It takes the trailing 30-day trajectory of that vector and measures its RMSE similarity against the windows that preceded six labeled historical episodes: the September 2019 repo spike, March 2020, the March 2023 SVB run, and the April, September, and December 2025 events.

Current output: the closest match is the run-up to the **September 2025 tax-date squeeze** at 0.58 similarity, aligned to one day before that break. **Interpretation discipline:** resemblance is context, not evidence. The Echo output is displayed alongside the index but deliberately carries zero weight in it. This is a documented editorial decision, changeable in one line of config.

4.5 RV X-Ray — how big is the leveraged Treasury complex, really?

Published basis-trade estimates disagree by \$500B because their methods are opaque. Seiche's method is deliberately simple and fully published:

```
pair proxy =  $\sum$  over 6 UST futures contracts of min(leveraged-fund shorts, asset-manager  
longs)  $\times$  face value  
DV01 =  $\sum$  shorts  $\times$  per-contract dollar-value-of-1bp (constants in config, tunable)
```

Current output: pair proxy **\$1,060B**, sitting independently on top of the IMF's ~\$1T estimate; aggregate DV01 \$557M per basis point; position down \$167B over 13 weeks (the complex is shrinking, not building); size z-score +0.49 against its own history. The shock table answers the question a risk desk actually asks: a 30bp adverse move marks the complex down \$16.7B, and an assumed 10% forced unwind (\$106B) would consume about 3% of one day's DVP repo volume (\$3,282B/day). The funding market can currently absorb an orderly unwind; the danger scenario is a disorderly one on a crunch date, which is why this engine is read together with Liquidity Weather.

4.6 Auction Digestion — is the market choking on Treasury supply?

Each Treasury auction is scored against the trailing distribution of its own tenor (minimum eight prior auctions): bid-to-cover z-score (weight 0.5, inverted), primary-dealer takedown share z (0.3, dealers as buyers of last resort), indirect-bidder share z (0.2, inverted, foreign demand). Bills are down-weighted 0.35 since coupons carry the duration-absorption signal. An EWMA(20) across auctions gives a cumulative indigestion index.

Current output: digestion index 0.16, mildly positive; the July 6 13-week bill scored 1.15 (weak) while surrounding auctions were fine. Nothing resembling the tailed 2s/5s/7s cluster of March 2026, which is this engine's reference signature.

4.7 Seiche Index — one number, fully decomposed, fail-loud

```
weights: tails .24 · kink .18 · weather .16 · rvxray .16 · srf .12 · auctions .08 · buffers .06
regimes: <25 CALM · 25–45 EROSION · 45–70 STRAIN · ≥70 STRESS
```

The composite is a weighted blend of engine sub-scores plus two level gauges: SRF usage (the "confession channel", since paying the ceiling rate is an admission that no cheaper private funding existed) and RRP buffer emptiness. Two properties matter more than the formula:

- **Full decomposition.** Every point of the index is attributable to a named component on screen. No black box.
- **Fail-loud.** If an upstream feed dies, its component renders as DEAD, its weight is renormalized away, and the published coverage percentage drops. A broken input can never silently make the index look calm. This was exercised for real during the build: when FRED throttled, the dashboard showed the degradation explicitly.

Component	Score now	Reading
buffers (RRP emptiness)	99.5	Shock absorber gone; structural fact of the regime
weather (forward path)	66	Crunch flags at Jul-31 and Aug-31 month-ends
kink (scarcity proximity)	55	Below a shallow kink; depth-scaled, consistency-checked
rvxray (positioning)	40	Large but shrinking complex
auctions (supply digestion)	32	Mild indigestion, no coupon stress
tails (acute stress)	14	Quiet
srf (confession channel)	1	Zero take-up
Seiche Index	38.8	EROSION · coverage 100%

5 · Data sources and architecture

Source	Agency	What Seiche pulls	Cadence / lag	Auth
FRED fredgraph.csv	St. Louis Fed	WALCL, WRESBAL, WTREGEN, RRPONTSYD, IORB/IOER, EFR, SOFR, GDP	Daily/weekly, T+1 to T+7	None
Markets API	NY Fed	SOFR/TGCR/BGCR with full percentile distributions; Standing Repo Facility operation results	Daily ~8am ET; ops same-day	None
Short-Term Funding Monitor API	Office of Financial Research	DVP and tri-party repo volumes and rates; TGCR/BGCR series; MMF holdings	Daily, T+1 preliminary	None
FiscalData API	US Treasury	Daily TGA opening balance; full auction results (bid-to-cover, dealer/indirect takedown); upcoming auctions	Daily ~4pm ET	None
Public Reporting (Socrata)	CFTC	Traders in Financial Futures: leveraged-fund and asset-manager positions in all six UST futures	Weekly, Friday for Tuesday (T+3)	None

System design

- **Backend:** Python 3.14, FastAPI on port 8787. One collector per agency with per-series fault isolation (a dead series degrades to its stale cached copy with its true staleness class, or to an explicit fault entry; it never kills the batch). SQLite cache with cadence-aware TTLs; warm snapshot assembles in ~3 seconds.
- **Frontend:** Vite + React + TypeScript + uPlot, dark terminal aesthetic, served by the same process in production. Provenance table renders every input with source, as-of date, fetch time, and a staleness chip (fresh / aging / stale / dead).
- **The editorial layer is quarantined.** Every judgment call (composite weights, regime thresholds, crunch cushions, the episode library, contract DV01 constants) lives in one file, `backend/seiche/config.py`, so the operator can tune the tool's voice without touching engine math.
- **Design lineage:** the provenance-native, fail-loud, confidence-native principles come from the Tiktó design system.

Battle scars from live verification (all fixed, all documented)

- FRED's CDN silently blackholes custom User-Agent strings (the connection hangs with no error) and tarpits browser UAs sent over Python TLS. The collector deliberately sends no UA header. This took three live test cycles to isolate.
- TGCR and BGCR return 404 on FRED's CSV endpoint; they are sourced from OFR instead.
- Quarterly GDP publishes with a lag; naive alignment silently truncated the kink fit at GDP's last observation (January), making the engine six months stale. Caught by reading the as-of date on the dashboard during visual verification.
- OFR volume mnemonics are denominated in raw dollars, not billions; the DVP figure briefly displayed as \$3.28 trillion billion per day.
- CFTC's TFF dataset drops the `_all` suffix on positioning fields; NY Fed's `last/N` rates endpoint rejects long windows and was replaced with explicit date-range search.

6 · Honest limitations

- **Daily cadence, not intraday.** The free stack's honest granularity is daily prints plus same-day operation results. Seiche will flag the run-up to a squeeze; it will not tick during one.
- **Positioning is T+3 and futures-only.** CFTC data cannot see repo-financed cash positions directly; the RV X-Ray is a proxy with a published method, not a census. Its value is trend and shock arithmetic, not the third decimal.
- **The kink is an estimate.** R^2 0.62 on a regime-spanning sample; the consistency check exists precisely because breakpoint estimates are unstable. Treat the distance-to-kink as a band, not a line.
- **Composite weights are editorial.** They encode the view that tails and spreads lead, structure amplifies, and positioning is the payload. A different desk could defensibly weight differently; the config file invites it.
- **No alerting yet** (designed for, not built), no swap-spread levels (needs paid data; the funding leg is proxied instead), single-user, local-first.
- **Not investment advice.** It is an instrument panel, not a trade recommendation engine.

7 · Roadmap

1. **Alerting** — webhook/email/Telegram on regime transitions, tail-z thresholds, SRF take-up > 0 , new crunch windows. The payload structure already supports it.
2. **Auction calendar overlay** — merge FiscalData's upcoming-auctions feed into Liquidity Weather so crunch windows show the specific settlements that cause them.
3. **Primary dealer positions** (NY Fed FR2004) as a dealer balance-sheet constraint gauge, and MMF flow decomposition from the OFR MMF dataset.
4. **Echo library growth** — each new stress episode gets labeled and joins the fingerprint library, making the engine smarter with every cycle.

5. **Deployment** — single-process design deploys to any small VPS; a public read-only instance with a premium alert tier is the natural freemium shape if this ever becomes more than a side project.

Seiche v0.1 · report generated 2026-07-06 · All input data from official US government APIs (Federal Reserve Board via FRED, Federal Reserve Bank of New York, Office of Financial Research, US Treasury, CFTC), verified against primary sources on the date of this report. Engine outputs are computed estimates with published methods. Not investment advice.